

AI-Driven Innovations to Enhance Tea Production and Quality in Sri Lanka's Mid-Country

Project ID: 25-26J-133

Project Proposal Report

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Sri Lanka's tea industry has been a cornerstone of the national economy since the 1800s and remains globally recognized for its quality. However, growing competition, fluctuating demand, and uncertain auction dynamics present challenges for producers and exporters in maintaining profitability. To address these issues, this research proposes the development of weekly tea price prediction models across different elevations and tea grades using advanced machine learning techniques. In addition, the study focuses on weekly auction volume forecasting for selected Mid-Country tea factories, incorporating key external factors such as production variability, demand patterns, and rainfall. Furthermore, a Smart Auction Behavior System is designed using a Multi-Agent System (MAS) framework to simulate the interactions among factories, brokers, and buyers in a dynamic auction environment. Data from multiple sources will be combined into a unified dataset to enhance model reliability and practical usability. The anticipated outcome is a forecasting and simulation framework that supports informed decision-making, reduces risks, and improves efficiency within the Colombo Tea Auction.

Keywords: Machine Learning, SARIMAX, Random Forest, Volume Forecasting, Multi-Agent System, Smart Auction, Tea Industry

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List of Abbreviations

Abbreviation	Description
XAI	Explainable Artificial Intelligence
AI	Artificial Intelligence
ML	Machine Learning
LSTM	Long Short-Term Memory
API	Application Programming Interface
RF	Random Forest
GB	Gradient Boosting
ANN	Artificial Neural Network
ARIMA	AutoRegressive Integrated Moving Average
MAS	Multi-Agent System
SARIMA	Seasonal Auto-Regressive Integrated Moving Average

Table 1: List of Abbreviations

1. INTRODUCTION

1.1 Background & Literature survey

Sri Lankan tea, globally known as Ceylon tea, is renowned for its high quality and distinctive flavor. As the third-largest tea exporter worldwide, Sri Lanka achieved a record tea export revenue of Rs.428 billion in 2023, surpassing the previous high of Rs.411 billion in 2022. In 2024, the country exported approximately 245.79 million kilograms of tea, generating around \$1.43 billion in revenue.[1] The tea plantations are primarily situated in the central mountainous regions of the country, divided into distinct elevation categories high-grown, mid-grown, and low-grown each producing teas with unique taste profiles and price ranges. While the majority of tea is produced in these highlands, some plantations exist at lower elevations as well.

The majority of Sri Lankan tea is sold through the Colombo Tea Auction, which has operated since 1883 and is held up to twice weekly.[2] Only a small percentage of tea is sold separately, such as premium single-estate teas that are packaged and marketed similarly to single malt whiskies, prized for their unique origin. At the auction, tea lots are identified by their grade and estate of origin. Each lot is tested by auction brokers who receive samples one week before the auction date; buyers then evaluate these samples to decide which lots to bid on.

Sri Lanka's tea plantations are distinctly categorized by elevation, which significantly determines the flavor, aroma, and overall quality of the tea produced. The key tea-growing regions include Kandy (medium-grown teas at approximately 2000–4000 feet), Nuwara Eliya (high-grown teas around 6000 feet), Dimbula and Uda Pussellawa (high-grown areas with elevations between 5000–6000 feet), and Ruhuna (low-grown teas mostly at sea level). Each region's unique altitude and climatic conditions contribute to the diverse characteristics and market values of the teas harvested there [3].

The variation in altitude also influences cultivation practices, yield levels, and market demand across different tea-growing areas. The smallholder sector, particularly in districts such as Matara, plays a critical role in national production, yet their practices and adoption of technology differ according to the elevation at which cultivation takes place. As highlighted by Al Sandika (2018) [18], the dissemination of technological information and advisory services to tea smallholders must therefore be tailored to elevation-specific needs, since production techniques, yield potential, and consumer preferences vary considerably across high, mid, and low-grown regions. To illustrate this, the elevation categories of tea cultivation in Sri Lanka are presented in Table 2, and the geographical distribution of tea-growing regions is shown in Figure 1.

Category	Altitude	Tea land extent [Hectare]	Total Production [million kgs]
High Grown	Above 1,200m	41137	76.9
Mid Grown	600 – 1,200 m	71018	49.7
Low Grown	Below 600m	109814	202.3
Total	-	221,969	328.9

Table 2: *Elevation Categories of Tea Cultivation in Sri Lanka-2015*[18]

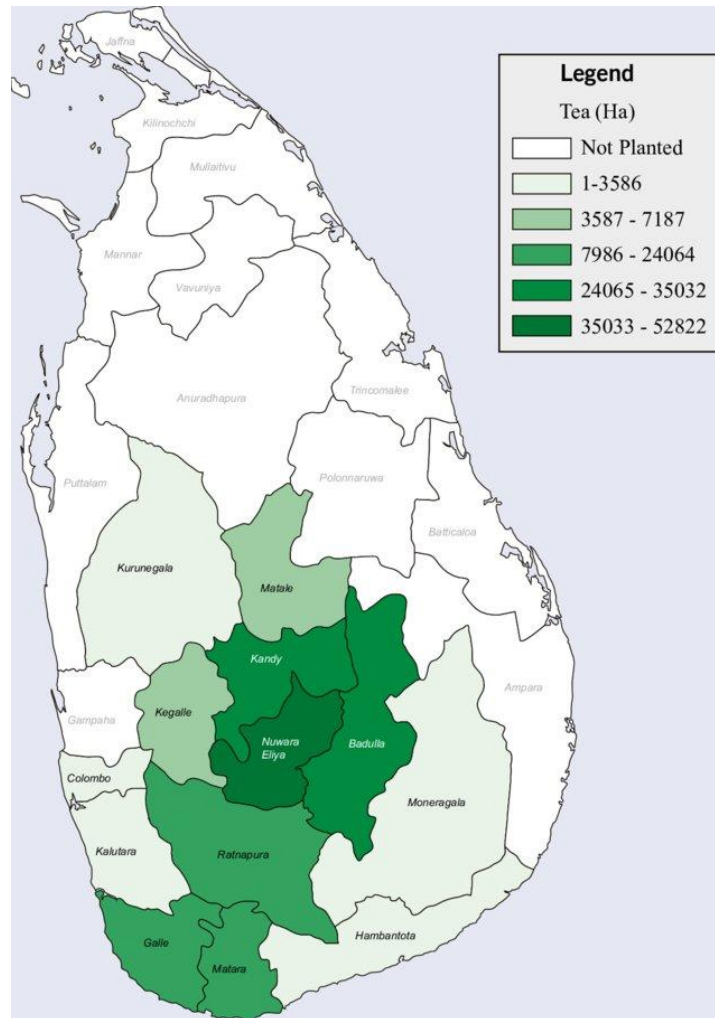


Figure 1.1: Tea Growing Regions Map

Tea price fluctuations at the auction depend on multiple factors, including supply volume, weather conditions, elevation-related characteristics, tea grade, festival demand, and macroeconomic variables such as exchange rates. Supply is a major determinant of price, with weather and climate predictions playing a crucial role in agricultural decisions.[7] As demand varies weekly, tea buyers highly value timely and precise price forecasts that factor in these seasonal and regional influences. Weekly prediction of tea prices by grade and elevation offers a significant competitive advantage, enabling buyers to optimize purchasing strategies.

Tea buyers seek this study to enhance their competitiveness by obtaining weekly price predictions based on tea grade and elevation, which provides a significant market advantage. Knowing the likely tea price enables buyers to make informed decisions

on whether to purchase at the upcoming auction or wait for a better opportunity. Predictive analytics benefits the entire agricultural supply chain by improving price transparency and aiding in optimal product selection. Ultimately, accurate price forecasting supports strategic planning and market efficiency for all stakeholders.[4]

Prior research on price prediction for the Colombo Tea Auction has largely relied on artificial intelligence and time series analysis to model auction price behavior.[6] These studies made important contributions by identifying the role of seasonality, demand shifts during holidays, and climate-related cycles in influencing tea prices. For example, Induruwage et al. (2016) emphasized capturing common seasonal patterns for monthly forecasts, providing insights into long-term demand fluctuations.[5] However, such monthly-based approaches, while effective for identifying broader market trends, lack the granularity required to capture rapid week-to-week dynamics that directly affect auction decisions. Since the Colombo Tea Auction operates twice weekly, buyers and sellers require high-frequency forecasts that align with this schedule. The absence of weekly-level modeling in prior work therefore represents a critical research gap one that this study seeks to address by advancing from broad, trend-oriented prediction frameworks toward fine-grained, operationally relevant weekly forecasting.

Recent advancements in tea price forecasting have been demonstrated by Ariyaratne et al. (2024), who introduced an automated weekly prediction model focused on the Broken Orange Pekoe Fannings (BOPF) grade across Uva High & Mid and Western High & Mid elevations. Unlike earlier studies that concentrated primarily on monthly price forecasting and broad seasonal patterns,[5][6] their work aligned directly with the Colombo Tea Auction's twice-weekly schedule, thereby increasing its operational relevance for buyers and brokers. A key methodological improvement was the adoption of SARIMAX models, a refined extension of Box-Jenkins approaches, which incorporated external variables such as supply, demand, rainfall, and the USD exchange rate. These exogenous factors enabled the model to capture localized auction dynamics more effectively than earlier models that treated prices largely as an outcome of internal time-series trends. The results, which achieved remarkable

predictive accuracy with Mean Absolute Percentage Errors (MAPE) as low as 0.26% for Western High elevation, illustrate how elevation-specific modeling and the integration of climatic and macroeconomic indicators can significantly outperform traditional monthly forecasting methods. This advancement provides an important foundation, yet its grade-specific and region-limited scope leaves room for further expansion to multi-grade and multi-agent frameworks, as pursued in the present study. [14]

Another significant contribution of Ariyaratne et al. (2024) was the discovery of unexpected inverse relationships between supply, demand, and auction prices across different elevations—contradicting the classical supply-demand assumption that prices rise when demand increases and fall when supply expands. Earlier studies on tea price forecasting[5][6] largely treated demand and supply as straightforward determinants, overlooking such elevation-specific anomalies. The methodological strength of Ariyaratne’s study lay in its systematic integration of estate-level weekly auction prices with meteorological and economic data, alongside practical features such as a CSV output function to improve end-user adoption. Compared to earlier models that were primarily academic and trend-focused, this approach demonstrated a stronger application-oriented design. By providing an elevation- and grade-specific, high-frequency prediction framework, the study advanced forecasting precision and practical usability. However, while it underscored the importance of incorporating multi-factor, seasonal, and exogenous variables, its scope remained limited to predictive modeling without addressing interactive market behaviors. Building on this foundation, our research extends beyond predictive accuracy to integrate multi-agent auction simulations, thereby bridging the gap between price forecasting and behavioral modeling for real-time decision support in Sri Lanka’s tea sector.

Earlier research on Sri Lanka’s tea sector has also focused on production forecasting rather than auction price dynamics. Kumarasinghe and Peiris, for instance, applied the Box-Jenkins ARIMA methodology to model annual tea production from 1964 to 2015 across the three major elevation zones: low-grown (up to 600 m), mid-grown (600–1200 m), and high-grown (above 1200 m).[13] Their study identified distinct

ARIMA configurations for each elevation ARIMA (2,2,1) for total and low-grown production, ARIMA (1,2,1) for mid-grown, and ARIMA (2,1,0) for high-grown and projected increases in production of 5.15%, 2.6%, and 4.5% respectively, leading to an overall 4.08% rise in national output by 2020. While these findings demonstrated the utility of ARIMA models in capturing long-term production trends shaped by climatic and agronomic conditions, the reliance on annual data limited the ability to capture short-term fluctuations in factory output and auction volumes. In contrast to such macro-level, production-focused studies, the present research shifts attention to weekly forecasting of both price and volume, thereby aligning more closely with the fast-moving dynamics of the Colombo Tea Auction and the decision-making needs of stakeholders at the factory and buyer level.

The foundational contributions of production-focused studies underscore the value of time series modeling for capturing long-term trends shaped by climatic, agronomic, and socio-economic conditions. However, their reliance on annual data and emphasis on production volume overlooks the short-term fluctuations and market-sensitive variables that directly influence auction outcomes. This creates a gap in addressing the week-to-week dynamics of both price and volume that are critical for factory managers and regional stakeholders. Building on these earlier insights, the present research advances beyond long-horizon production forecasting by adopting a weekly granularity and explicitly integrating price predictions with production volume estimates. Furthermore, by embedding a multi-agent simulation of auction behavior, our framework extends traditional statistical approaches into a more operational, decision-oriented tool. This not only improves the practical alignment with the Mid-Country tea sector's needs but also positions the study to bridge the gap between macro-level forecasting and micro-level auction decision support.

To address the diverse climatic and geographical characteristics of Sri Lanka's tea-growing regions, this research develops three separate price prediction models, each tailored specifically for one of the key elevation-based regions: Low, Mid and High regions. Unlike many previous studies that apply a single generic forecasting model

across all regions, our approach recognizes that regional and elevation-driven variations significantly impact tea quality, market behavior, and price dynamics. By localizing the price prediction models, we aim to capture region-specific trends and seasonal patterns more accurately, thereby enhancing forecast precision and practical utility for stakeholders in each zone.

Sri Lanka's Mid-Country tea-producing region, situated between 600 m and 1,200 m in elevation, holds a distinctive position within the national industry by producing medium-grown teas characterized by bright liquor and strong flavor. While high- and low-grown teas have often dominated both research and market discussions due to their export prominence, the Mid-Country has received comparatively less scholarly and policy attention despite its strategic role in balancing quality and volume in the Colombo Tea Auction. The region, concentrated primarily in the Central Province's Kandy and Matale districts, hosts 28 registered factories (27 in Kandy and 1 in Matale), which act as critical nodes in sustaining supply chain stability and ensuring consistent quality standards. Unlike low-grown regions that are more smallholder-intensive, or high-grown regions that emphasize premium branding, Mid-Country factories operate in a hybrid space providing both large-scale processing capacity and essential income generation for smallholders. This comparative underrepresentation in prior research reinforces the importance of focusing on Mid-Country auction dynamics in the present study, particularly given its contribution to regional employment, smallholder livelihoods, and export competitiveness.

Factory Code	Factory Name	Location
MF1024	Allen Valley Tea Factory	Morawakorale Co-Operative Society Ltd, Deniyaya Road, Kotapola
MF0141	Alma Tea Factory	Kandapola
MF0077	ASN Malgolla Tea Factory (Pvt) Ltd	Dolosbage, Nawalapitiya
MF0402	Elpitiya Tea Factory	Weligalla, Gampola
MF1461	Embilmeegama Tea Factory	No.151, Colombo Road, Embimeegama, Pilimathalawa
MF0967	Gatagahawela Tea Factory	Ranagala, Makuldeniya, Theldeniya
MF0251	Glenloch Tea Factory	Nuwara Eliya Road, Katukitula
MF1044	Greenwood Tea Factory	Greenwood Estate "A", Nawalapitiya
MF0835	Harangalla Tea Factory	Harangalla Plantation (Pvt) Ltd, Kataboola, Nawalapitiya, Kataboola
MF0722	Hatale Tea Factory	Meezan Hatale Plantations (Pvt) Ltd, Hatale, Panwila
MF1522	James Valley Tea Factory	Nawa Gurukele, Doluwa
MF0864	Mahavila Tea Factory	Ulapane
BF0171	New Fernland Tea Factory	Ulapane, Gampola
MF0150	New Orayan Tea Factory	Orayan Estate, Gampola
BF0043	New Weliganga Tea Factory	Nawalapitiya Road, Ulapane
BF0170	Nildalukanda Tea Factory	Kadugannawa Road, Welamboda, Gampola
MF1181	Pallegama Tea Factory	Pallegama Estate, Ulapane
MF1082	Raja-Ela Tea Factory	Railway Station, Raja-Ela, Ulapane
MF0148	Raxawa Tea Factory	Panwila , Kandy
BF0132	Roslin Hill Tea Factory	Hindagala, Peradeniya

MF1512	Rozella Tea Factory	Binoya 'A' Estate , Rozella
MF1523	Udayakanda Tea Factory	184/B, Kandy Road, Welamboda
MF1257	Uplands Tea Factory	Upland Estate, Peradeniya
BF0020	Watadeniya (Hameedia) Tea Factory	Watadeniya, Welamboda
MF1582	Kadugannawa Tea Factory	167, Colombo Rd, Kadugnnawa
MF0127	Kings Valley Tea Factory	NuwaraEliya road, Atabage
MF0020	Kurugama Tea Factory	Murutalawa, Via Kandy
MF1412	Lower Donside Tea Factory	Nawalapitiya
BF0166	Mahatenna Tea Factory	2/B, Mahatenna Estate, Yahalatenna, Kandy
MF0093	Ancoombra Tea Factory	Meezan Ancoombra Plantations (Pvt) Ltd 392, Main Street, Matale

Table 1.2: List of Mid-Country Tea Factories in Central Province

Each factory produces a range of tea types and grades reflecting the unique characteristics of their specific elevation zones. Given the variable production volumes and quality across these factories, this research emphasizes forecasting auction volumes and prices at the factory level to better support localized decision-making for growers and factory managers.

In terms of volume, approximately 97% of tea produced by factories is sent to auction, with the remaining 3% sold directly at the factory level. Each Mid-Country tea factory dispatches varying volumes to the auction weekly, which underscores the need for accurate volume forecasting to assist factory managers and stakeholders in production planning and market timing.

Recent advances in regional tea yield forecasting demonstrate the growing effectiveness of machine learning in agronomically diverse environments. For instance, a 2023 study in Bangladesh analyzed more than 50 years of production and weather data across 23 tea-growing regions, applying Random Forest, Gradient

Boosting, and ensemble methods, with Random Forest achieving the highest accuracy of 97% [9]. This underscores the importance of integrating long-term, multi-source datasets with climatic variables such as rainfall, humidity, and temperature to capture seasonal variability in agricultural productivity. Complementing these yield-focused approaches, recent research in volume forecasting has highlighted the potential of hybrid models that combine traditional time series with machine learning.

Zhao (2023), developed a hybrid ARIMA Random Forest framework to predict cargo volumes in e-commerce sorting centers, effectively modeling temporal features such as trend, seasonality, and lag effects. Their rolling forecast mechanism demonstrated superior adaptability to real-time market fluctuations, with Random Forest outperforming neural networks, SVMs, and regression baselines on MAE and RMSE metrics. Taken together, these studies illustrate how blending domain-specific time series methods with feature-rich machine learning enhances prediction accuracy in complex, dynamic environments. Extending this logic to Sri Lanka's tea sector, where weekly auction volumes fluctuate under both climatic and market pressures, hybrid forecasting frameworks can provide more robust, factory-level decision support an approach advanced in the present study.[15]

These findings from regional and cross-country studies are particularly relevant to Sri Lanka's tea sector, which exhibits comparable climatic diversity and elevation-driven production variability. Prior Sri Lanka-specific research has demonstrated that elevation, grade, and seasonal factors significantly affect both auction prices and production volumes [10]. While these studies established important correlations and highlighted key predictive variables, they often relied on limited datasets or traditional modeling approaches that do not fully capture high-frequency, weekly fluctuations at the factory or auction level. By adopting similar machine learning methodologies, supported by comprehensive environmental and historical datasets, the present study extends these approaches to generate precise, region-specific forecasts that integrate both supply and market dynamics. This advancement not only enhances the accuracy of auction price and volume predictions but also strengthens decision-support capabilities for smallholders, factory managers, and market planners, providing a more operationally actionable framework than prior work.

An important and innovative aspect of this research is the development of a Smart Auction Behavior Simulation leveraging multi-agent Artificial Intelligence. This simulation models the interactions among market players including factories, brokers, and buyers capturing the complex dynamics of auction supply and demand, competition intensity, and buyer behavior.

Multi-Agent Systems (MAS) have been successfully applied in diverse domains where human behavior, decision-making, and interactions play a critical role. For example, Aguado (2018) developed an MAS framework for social networks that integrates sentiment and stress analyzers to predict and mitigate negative outcomes in online interactions. Their system employs specialized agents such as sentiment analyzers, stress analyzers, and a combined analyzer that collaboratively process social media text to provide advice or warnings to users before posting, thereby supporting decision-making in emotionally high-risk situations. This demonstrates how MAS architectures, structured into layered components (presentation, logic, and persistence), can effectively model complex socio-technical environments by combining analytical agents with advisory mechanisms. Such applications reinforce the suitability of MAS for auction markets like Sri Lanka's tea sector, where different stakeholder agents (factories, buyers, and brokers) interact dynamically, and decision-making can be guided by predictive and advisory components [17].

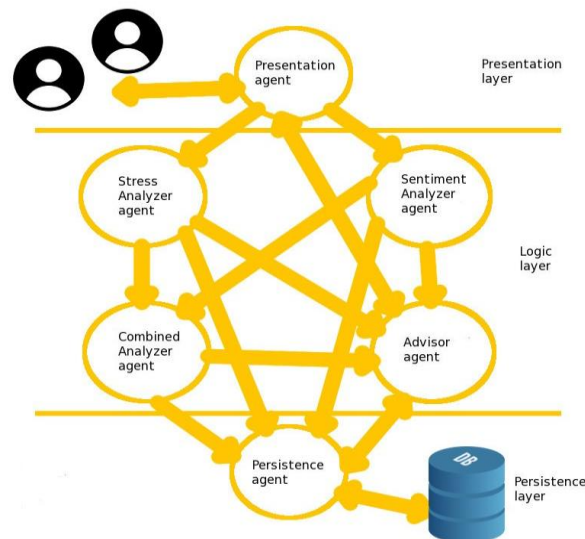


Figure 1.2. Architecture of the system built as a Multi-Agent System (MAS) with three layers.

Figure 1.2 illustrates the overall architecture of the Multi-Agent System (MAS), highlighting the interactions among different agent types across the system's three layers. The Presentation Layer serves as the interface with users, where the Presentation Agent collects inputs such as auction participation intentions, factory data, or buyer preferences and displays relevant outputs including recommendations or warnings.

The Logic Layer contains the core processing agents responsible for auction dynamics. These include the Factory Agent, Broker Agent, and Buyer Agent, each designed to model real-world roles in the tea auction ecosystem.

- **Factory Agent:** Manages production and supply information for each tea factory. It receives inputs such as forecasted weekly volume, tea grade and elevation, and production cost along with minimum acceptable prices. The agent prepares lots for auction and coordinates with Broker Agents to ensure optimal lot allocation and pricing strategies.
- **Broker Agent:** Facilitates the auction process by handling lot capacities, commission rates, and analyzing historical price trends. This agent provides guidance to both Factory and Buyer Agents, helping optimize pricing, lot distribution, and auction efficiency.
- **Buyer Agent:** Represents auction participants looking to purchase tea. It uses budget constraints, price sensitivity, demand quantity, and bidding strategies to make informed auction decisions. The agent dynamically adjusts bidding behavior based on market conditions, competition, and recommendations from Broker Agents.

The Output Layer includes reporting and decision-support tools that provide actionable insights, combining auction forecasts, predicted volumes, and simulation results. This layered MAS design ensures modular, coordinated interactions among autonomous agents while capturing the complex dynamics of Sri Lanka's tea auction system. By integrating production, brokerage, and buyer behavior, the MAS enables

realistic simulation of market outcomes and supports strategic decision-making for all stakeholders. [17]

Agent-based simulation, also known as multiagent simulation, is a computational modeling approach that represents complex systems as interactions among autonomous entities called agents. Each agent acts independently according to its own rules, objectives, and decision-making processes, while also responding to changes in the environment and the behaviors of other agents. This decentralized framework captures the diversity and adaptability of real-world market participants, making it especially effective for modeling intricate systems like auctions, where different actors such as sellers, buyers, and brokers have distinct roles and strategies.

In the context of the Colombo Tea Auction, agent-based simulation enables the virtual recreation of the bidding process by modeling tea factories, brokers, and buyers as intelligent agents. These agents are programmed with attributes such as factory production capacity, buyer budget constraints, and broker negotiation tactics. The simulation iterates on a weekly basis, reflecting the auction's schedule, allowing agents to dynamically adjust their supply, bids, and strategies based on market conditions and the observed behavior of competitors. This dynamic interaction creates an emergent market environment where auction prices, volumes, and allocation outcomes evolve naturally—providing a realistic picture that cannot be fully captured through traditional price or volume forecasting models alone.

Unlike conventional forecasting methods that focus solely on predicting prices or volumes, agent-based simulation emphasizes the behavioral and strategic dimensions of auction markets. It helps reveal how factors like competition intensity, bidder behaviors, cooperation among agents, and stochastic uncertainties impact auction fairness, efficiency, and risk. By simulating various “what-if” scenarios such as sudden supply shocks, entry of new buyers, or changes in auction rules stakeholders can evaluate potential outcomes before making real-world decisions. This capability is crucial for an industry like Sri Lanka's tea sector, where market dynamics are influenced by a complex interplay of production variability, buyer preferences, and regulatory mechanisms.

Integrating multiagent simulation with precise weekly price and volume forecasts represents a significant advancement in tea auction. This combined approach offers a comprehensive decision-support tool that not only anticipates market trends but also actively models the market's interactive behaviors. As a result, it empowers tea producers, brokers, and buyers with insights that enhance strategic planning, mitigate risks such as oversupply and price volatility, and improve overall market transparency and competitiveness. Such an innovative framework aligns well with the needs of Sri Lanka's Mid-Country tea sector, fostering a data-driven and adaptive approach to auction participation and market development.

Multi-agent simulation has been extensively applied to study complex systems where multiple autonomous agents interact, compete, and cooperate to achieve individual or collective objectives [8][11]. In auction-based contexts, these frameworks have demonstrated their effectiveness in coordinating agents under uncertainty and resource constraints. For instance, Tang et al. (2016) employed a distributed auction mechanism combined with improved Truncated Lévy walks to model search-and-rescue teams in landslide scenarios, showing that cooperative auction-based strategies outperformed non-cooperative approaches in efficiency, robustness, and resource utilization [11]. Similarly, Ebata and Ariyoshi (2023) highlighted the capacity of multi-agent simulations to capture complex social dynamics and predict outcomes in systems with interacting agents [8]. While these studies validate the effectiveness of multi-agent frameworks in non-agricultural or emergency-management settings, their application to agricultural markets particularly auction-based commodity markets like Sri Lanka's tea sector remains limited. The present research builds on these insights by adapting multi-agent auction simulation techniques to model the dynamic behaviors of tea factories, brokers, and buyers, thereby extending prior methodological advances to a novel, market-oriented agricultural context.[8].

By adopting these principles, our Smart Auction Behavior Simulation allows stakeholders in the tea market to test “what-if” scenarios, anticipate auction risks such as oversupply or low competition, and evaluate optimal bidding and selling strategies. Each agent in the simulation represents a factory, broker, or buyer, with behavior rules

influenced by historical auction data, competitive pressures, and market conditions. The simulation also incorporates stochastic decision-making to model real-world uncertainties in supply, demand, and pricing. By integrating these features, this research provides a data-driven decision-support tool that enhances operational planning, mitigates auction risks, and improves market efficiency in Sri Lanka's tea industry.

In addition to forecasting price and volume, understanding the behavioral interactions among auction participants during bid evaluations is essential for improving auction fairness and efficiency. Liu et al. (2013) studied bid evaluation in grouped multi attribute reverse auctions (GMAA), a common procurement method where bids are assessed not only on price but also on quality, service, and technical attributes. Their multiagent-based simulation revealed how antagonistic feelings between business and technical expert referee groups, influenced by bounded rationality and differing objectives, can grow over successive auctions. This dynamic may reduce the objectivity of bid evaluations, occasionally causing the loss of the best bids and compromising auction fairness.[16]

The multiagent approach models autonomous agents representing bidders, auctioneers, and referee groups, encapsulating their decision rules and interactions. It highlights the non-linear, time-varying behaviors inherent in complex auction systems through repeated simulations, exposing inefficiencies in current evaluation mechanisms. Such insights are invaluable for markets like the Colombo Tea Auction, where multiple buyer and seller agents interact competitively with bid evaluations influenced by subjective factors like quality grading and estate reputation. By integrating this behavioral simulation knowledge with predictive analytics of prices and volumes, our Smart Auction Behavior Simulation will not only forecast market outcomes but also dynamically model how participant behaviors and evaluation processes shape auction efficiency. This dual approach addresses a gap in current tea market literature, which largely overlooks the intricacies of auction evaluation mechanisms and their impact on market fairness and pricing strategies.

This research focuses specifically on the Mid-Country tea sector and aims to address three major objectives, identified in consultation with industry professionals, including tea producers, tasters, and auctioneers:

- Implement a weekly tea price prediction model for selected tea grades across different elevation categories, incorporating key factors such as historical auction prices, rainfall, exchange rates, and festival periods.
- Develop a weekly volume prediction model for selected Mid-Country tea factories and tea types, based on historical auction volume data and production characteristics.
- Implement a Smart Auction Behavior Simulation leveraging multi-agent AI frameworks to provide realistic modeling of auction dynamics, improve risk assessment, and inform bidding and selling strategies.

By concentrating on weekly timelines and integrating auction price, volume data, and simulated auction scenarios, this study leverages advanced AI and machine learning techniques to provide actionable market forecasts and decision support. The combination of elevation-aware price modeling, factory-level volume prediction, and intelligent auction simulations aims to empower stakeholders with data-driven insights to optimize selling decisions, improve market efficiency, and enhance competitiveness in Sri Lanka's vital tea industry.

1.2 Research Gap

In examining the existing literature on tea price forecasting, market demand prediction, and auction modeling, it is evident that most studies rely on generalized, single-model approaches [1][5][14]. These models often treat the tea industry as a uniform market, ignoring critical local variations such as elevation, tea grade, and regional climatic differences. Sri Lanka's tea-growing regions Low-grown, Mid-grown, and High-grown exhibit distinct production characteristics, quality profiles, and price behaviors. By relying on a single generic model, prior research often fails to account for these regional differences, reducing prediction accuracy and limiting practical usefulness for tea producers and brokers.

To overcome this limitation, the proposed research develops three separate predictive models, each tailored for a specific elevation zone: Low-grown (<600 m), Mid-grown (600–1200 m), and High-grown (>1200 m). Each model incorporates region-specific factors, including historical auction prices, local weather patterns, seasonal trends [1][5]. The Low-grown model focuses on Ruhuna and Sabaragamuwa regions, where global demand fluctuations significantly impact price. The Mid-grown model targets the Central Province and includes factory-level volume forecasting, a feature absent in previous studies [13]. The High-grown model emphasizes premium plantations in Nuwara Eliya and Dimbula, where scarcity and high demand make precise forecasting critical. Separating the models by elevation ensures localized, accurate predictions, improving decision-making for all stakeholders.

A major gap in prior research is the absence of factory-level volume prediction, particularly for the Mid-Country region. Most studies aggregate production at the regional or national level, which limits actionable insights for individual factories [13][14]. Factory-level predictions consider weekly dispatch volumes for each tea type (BOP, BOPF, Pekoe, Dust), based on historical production, factory capacity, regional demand, and competitive auction volumes. By providing tailored volume forecasts, this system enables factories to make informed decisions on whether to sell immediately or store tea, factoring in storage costs, potential price changes, and quality

degradation. This granular approach is unprecedented in Sri Lanka's tea research and directly links production planning with market strategy.

Another critical limitation in existing studies is the lack of dynamic auction behavior modeling. Traditional approaches treat auctions as static or purely statistical events [5][14][16]. The proposed research introduces multi-agent auction simulation, where each buyer, seller, and broker is modeled as an intelligent agent with distinct strategies. This simulation allows the system to detect oversupply risks, low-competition weeks, and optimal selling times. Furthermore, the integration of Explainable AI (XAI) techniques such as SHAP and LIME ensures that stakeholders understand the reasoning behind predictions and recommendations. Unlike prior studies, this approach provides actionable insights into auction dynamics, enhancing risk mitigation and decision-making capabilities.

Existing forecasting systems also overlook seasonal and festival-driven demand fluctuations, which strongly influence tea prices and auction volumes [1][5]. This research incorporates cultural and religious holidays Sinhala/Tamil NewYear, Christmas, Ramadan into the models, capturing demand spikes for specific tea grades before major celebrations. By doing so, the system provides more realistic forecasts, enabling producers and brokers to align production, storage, and auction timing with market demand. This ensures higher profitability and reduced unsold inventory, which traditional models fail to address.

Most past studies rely on statistical or machine learning methods such as regression, ARIMA, or simple neural networks for forecasting [5][6][14]. However, these approaches typically do not differentiate between elevation-specific trends or integrate multi-agent market simulations. By combining three elevation-specific predictive models, factory-level volume forecasting, festival-adjusted price adjustments, and multi-agent auction simulations, this research provides a holistic, end-to-end decision support system. This combination bridges significant gaps in the literature, offering both predictive accuracy and actionable insights for practical implementation in the Sri Lankan tea market.

The novelty of this research lies in its multi-layered, localized, and practical methodology. First, the separation of models by elevation captures regional climatic and market differences. Second, factory-level Mid-Country volume forecasting provides operational-level insights previously unexplored. Third, multi-agent auction simulation with XAI allows stakeholders to understand market dynamics and potential risks in real time. Finally, festival-based demand integration ensures that forecasts reflect real-world consumption patterns. Taken together, these features constitute a unique and innovative framework that addresses limitations of past studies and provides a robust tool for tea producers, brokers, and auction participants.

In conclusion, while prior research has contributed foundational knowledge in statistical and machine learning-based tea price prediction, it lacks localization, granularity, market simulation, and practical decision support components. This research addresses all these gaps by offering elevation-specific forecasting, factory-level volume predictions, festival-adjusted demand modeling, and dynamic auction simulations with transparent reasoning. As such, it represents a comprehensive, accurate, and practically applicable solution, directly enhancing profitability, risk management, and operational efficiency in Sri Lanka's Mid-Country tea sector [1][5][8][11][13][14][16].

Key Research Gaps:

1. **Generalized Forecasting Models:** Most existing studies adopt a single, generalized forecasting model for Sri Lanka's tea industry, without differentiating between Low, Mid, and High-grown regions. This overlooks elevation-based variations in production, quality, and market dynamics.
2. **Lack of Factory-Level Forecasting:** Current research focuses mainly on national or regional auction volumes and prices, but no studies attempt factory-level forecasting (particularly in the Mid-Country). This leaves a major gap in providing granular insights for producers and factory managers.
3. **Absence of Auction Behavior Simulation:** Previous work predicts tea prices or volumes in isolation, but does not simulate real auction dynamics. The lack of multi-agent auction simulations (modeling buyers, brokers, and sellers with

competitive strategies) means stakeholders cannot test strategies or anticipate market responses.

4. **Limited Use of Advanced AI and ML Techniques:** Existing studies rely heavily on traditional statistical methods (ARIMA, regression) or basic machine learning, which fail to capture complex interactions like climate effects, exchange rates, or seasonal/festival-driven demand.
5. **Weak Decision-Support Features:** Current models mainly provide raw forecasts (numerical outputs) with little practical guidance on selling, storing, or adjusting strategies. There is a clear need for predictive systems that also deliver actionable recommendations.

Table 1.3: Research gap

Feature	Existing Systems	Proposed System
Forecasting Scope	Use a single generalized model for the entire tea industry, without regional differentiation.	Develops three separate elevation-specific models (Low, Mid, High grown) for localized and accurate predictions.
Granularity of Prediction	Predictions are made at regional/national level, lacking factory-level insights.	Provides factory-level volume forecasting for Mid-Country factories, supporting operational decision-making..
Auction Modeling	Treat auctions as static price prediction events with limited market dynamics.	Implements multi-agent auction simulation, modeling brokers, buyers, and sellers with distinct strategies.

Transparency & Explainability	Most models act as black boxes, offering limited interpretability.	Integrates Explainable AI (XAI) methods to justify forecasts and auction recommendations.
Decision Support	Provides only numerical predictions with little practical guidance.	Delivers actionable insights whether to sell immediately, store tea, or adjust auction strategy.

1.3 Research Problem

Sri Lanka, globally renowned for its Ceylon tea, is one of the largest exporters of black tea in the world. The tea industry plays a crucial role in the national economy, providing both export revenue and employment for millions of people, especially in the Central Province. According to the Sri Lanka Tea Board, tea export earnings crossed Rs.428 billion in 2023, highlighting its economic importance. However, despite this significant contribution, the industry faces persistent challenges due to price fluctuations and unstable supply-demand conditions.

One of the most pressing issues lies in the Colombo Tea Auction, where tea is traded weekly. The prices determined at this auction directly influence both smallholder farmers and factory owners. When auction prices fall below expectations due to oversupply, farmers and factories struggle with reduced profit margins, sometimes even operating at a loss. Conversely, during undersupply periods, exporters and buyers face difficulties meeting demand, disrupting the market balance. These fluctuations have a cascading impact on decision-making throughout the supply chain, including crop planning, factory processing, and auction bidding strategies.

Another critical problem is that existing forecasting mechanisms are often generalized. Current approaches mainly focus on national-level price predictions or broad export figures, without addressing factory-level production volumes and elevation-based differences. The Sri Lankan tea industry is uniquely segmented into low-grown, mid-grown, and high-grown teas, each with its own quality characteristics and market demand. Among these, mid-country teas hold strategic importance due to their significant share in overall production. However, very limited research has been carried out on predicting factory-level auction volumes and prices specifically for mid-country regions. Without such focused forecasting, both growers and factory managers remain uncertain about production targets, resulting in inefficiencies and financial losses.

Moreover, the lack of data-driven decision-making tools further aggravates the problem. Farmers and factories often depend on historical experience, intuition, or

short-term trends, rather than systematic forecasting models. This results in overproduction during peak seasons, leading to oversupply at auctions and steep price drops. On the other hand, underestimation of demand may create scarcity, preventing exporters from meeting their orders and weakening Sri Lanka's competitive edge in the global tea market. These mismatches highlight the absence of robust, factory-level forecasting frameworks that can guide stakeholders in aligning production with market demand.

Therefore, there is a pressing need for a predictive system that can forecast both auction prices and factory-level production volumes, with a special focus on the mid-country region. Such a system would allow smallholders, factories, and buyers to anticipate market conditions more accurately, make informed decisions, and stabilize income levels. Addressing this problem not only benefits stakeholders in the supply chain but also strengthens the overall resilience of Sri Lanka's tea industry in the face of market fluctuations.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop an AI-powered web application that supports data-driven decision-making in Sri Lanka's tea industry, with a particular focus on mid-country tea factories. The system is designed to accurately forecast weekly tea auction prices and factory-level dispatch volumes by analyzing historical auction data, elevation-specific trends, seasonal demand fluctuations, and market dynamics. In addition, the project incorporates agent-based simulations to model smart auction behaviors, enabling stakeholders to anticipate market conditions, competition, and optimal selling strategies. By integrating Explainable AI (XAI), the system provides transparent reasoning behind its predictions and recommendations, allowing users to understand market risks and take informed actions. Ultimately, this research aims to create a robust decision-support platform that enhances operational efficiency, reduces uncertainty, and strengthens the competitiveness and profitability of Sri Lanka's tea sector.

2.2 Specific Objectives

- **Develop an AI-driven web application for tea market forecasting:**

This objective focuses on creating a comprehensive platform that integrates price prediction, factory-level volume estimation, and auction simulations to support decision-making for mid-country tea stakeholders.

- **Forecast weekly tea auction prices with elevation-specific trends:**

The system will use specialized machine learning models to capture price patterns for low-, mid-, and high-grown teas, incorporating historical auction data, seasonal effects, and weather conditions to enhance accuracy.

- **Predict factory-level tea volumes for mid-country factories:**

The platform will estimate weekly dispatch volumes by tea grade (BOP, BOPF, Pekoe, Dust) for each mid-country factory, considering historical production, factory capacity, and market demand fluctuations.

- **Simulate auction behaviors using agent-based models:**

Buyers, sellers, and brokers will be represented as autonomous agents in the simulation to study supply-demand dynamics, competitive bidding, and optimal selling strategies. Explainable AI (XAI) will provide transparency for the simulation's recommendations.

- **Offer an interactive and user-friendly web interface:**

The platform will visualize data via dashboards, charts, and 3D maps, allowing users to explore forecasts, auction risk indicators, and factory-specific insights easily.

- **Enable real-time updates and insights:**

The system will provide stakeholders with continuous updates, explanations for predictions, and market alerts, promoting trust in AI-driven recommendations and supporting timely decision-making.

3. METHODOLOGY

The methodology of this research focuses on developing a data-driven web application aimed at enhancing decision-making in Sri Lanka's mid-country tea auction system. The approach begins with a comprehensive literature review of challenges in the tea industry, including price fluctuations, unpredictable demand, auction inefficiencies, and limitations of current forecasting techniques. This review also examines recent technological advancements in price prediction, production volume forecasting, and agent-based simulations, in order to identify gaps that the proposed system intends to address once implemented.

In line with the proposed design, the web application will be structured to improve accuracy in weekly price and factory volume predictions while remaining user-friendly for stakeholders such as tea brokers, smallholder farmers, and factory managers. The planned system will rely on historical data including weekly auction prices, factory-level dispatch volumes, tea grades, buyer participation, and elevation-specific production details. If available, weekly weather data (rainfall, temperature) and festival-related demand indicators will also be incorporated to enhance model precision. The interface will be designed to present complex information clearly and provide actionable insights.

The first component of the planned system will be weekly tea price forecasting using Time Series models. These models are intended to capture elevation-specific trends, seasonal variations, and market dynamics by analyzing historical auction prices, climate information, and demand indicators. By separating forecasts for low-, mid-, and high-grown teas, the system aims to provide localized insights that generic models may overlook, improving predictive reliability for stakeholders once operational.

For factory-level volume prediction, the proposes a hybrid forecasting approach that integrates both statistical patterns and machine learning techniques. This method will make use of historical dispatch volumes, production capacities, and regional demand trends to generate accurate weekly forecasts of tea quantities for each mid-country factory. By leveraging these insights, factories will be able to plan supply more

effectively, minimize risks of overproduction or shortages, and better align their weekly outputs with prevailing market expectations once the system is implemented and trained with real-world data.

The third component is a multi-agent auction simulation. In the proposed design, agents representing buyers, sellers, and brokers will simulate auction interactions, competition intensity, and strategic bidding behavior. Explainable AI (XAI) will be integrated to ensure that the reasoning behind recommendations is transparent, allowing users to understand auction risks, optimal selling windows, and potential outcomes. The simulation will also consider stochastic elements to reflect real-world uncertainties in supply, demand, and pricing.

Once the models are fully trained and the simulation module is implemented, the system will provide actionable sell/store recommendations based on predicted prices, storage costs, and potential quality changes. Interactive dashboards, charts, and maps will visualize predicted trends, auction risk indicators, and factory-level insights, enabling stakeholders to explore forecasts and export reports for decision-making. This user-centered design will ensure accessibility for both large and small-scale actors in the tea supply chain.

The planned methodology emphasizes ethical handling of data, transparency in algorithmic recommendations, and scalability of the system. All collected data will be used exclusively for academic purposes and treated with confidentiality. Once implemented, the models will be continuously refined as new data becomes available to improve forecasting accuracy. The research aims to deliver a robust, adaptive, and explainable decision-support platform that can enhance transparency, optimize production planning, and improve profitability for stakeholders in Sri Lanka's mid-country tea industry.

3.1 Project Overview

The project aims to develop a web-based decision-support system for the mid-country tea sector in Sri Lanka, integrating weekly price forecasting, factory-level volume prediction, and multi-agent auction simulation. By leveraging historical auction data, elevation-specific production details, climatic variables, and market demand indicators, the system provides stakeholders with actionable insights to optimize selling decisions and production planning. The project combines time series analysis, machine learning, and agent-based modeling to generate accurate, localized forecasts and simulate realistic auction behaviors. Interactive dashboards, charts, and visualizations are included to facilitate user understanding and support strategic decision-making for tea brokers, factory managers, and smallholder farmers. The overarching goal is to enhance transparency, improve operational efficiency, and support profitability across the mid-country tea supply chain.

3.2 System Overview Diagram

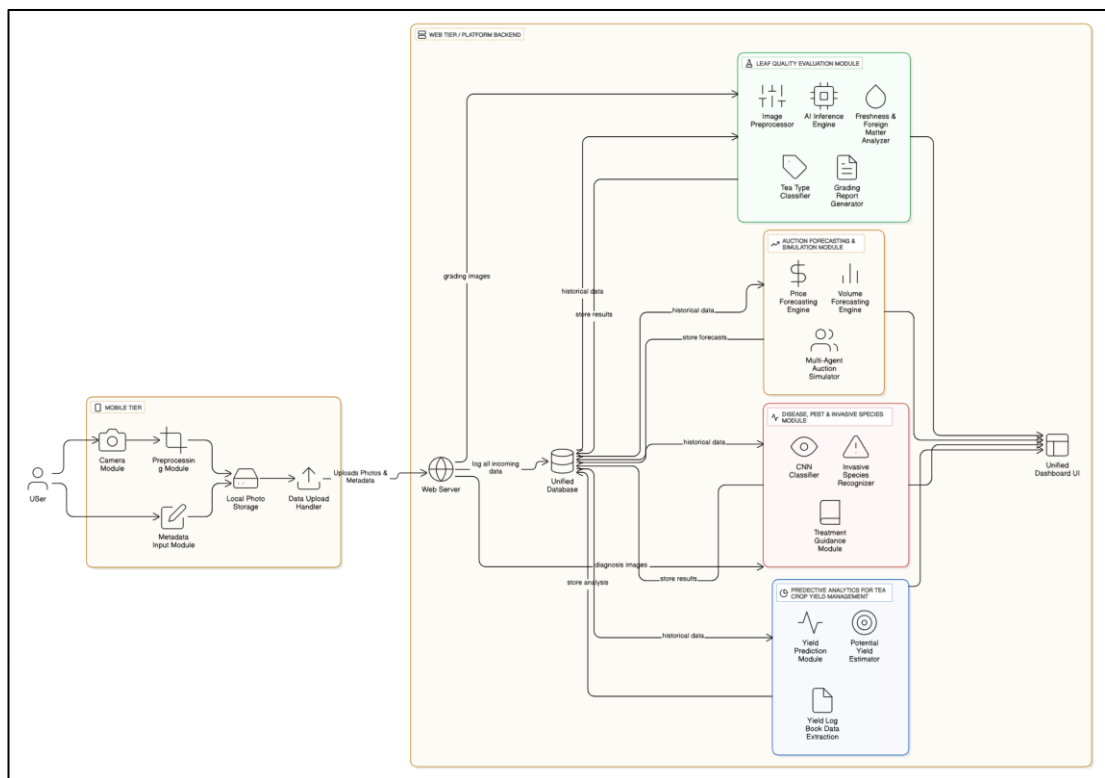


Figure 3.1: System Overview Diagram

3.3 System Overview

3.1 Project Overview The proposed system is designed to provide a comprehensive AI-powered solution for Sri Lanka's tea industry, addressing critical challenges such as declining yields, quality inconsistencies, labor shortages, and market inefficiencies. The system integrates four core modules: automated tea leaf grading, disease detection, yield prediction, and market demand forecasting into a unified decision-support platform accessible to tea estates, factories, and smallholder farmers. Figure 3.1 illustrates the overall system architecture. Among these modules, the tea leaf disease and pest detection component is the primary focus of this project. It employs deep learning techniques, particularly Convolutional Neural Networks (CNN), to accurately detect major tea leaf diseases such as blister blight, red rust, white spot, algal leaf spot, red leaf spot, anthracnose, and pink disease, as well as pest infestations including tea mosquito bug, red spider mite, tea tortrix, and shot hole borer. By analyzing high-resolution images collected from tea estates and leveraging real-world local data from the Tea Research Institute (TRI) of Sri Lanka, the system ensures early and precise identification of affected leaves. In addition to detection, the system provides treatment guidance, offering actionable suggestions based on expert knowledge to help farmers and estate managers take immediate corrective measures. A user-friendly dashboard consolidates disease alerts and treatment suggestions, enabling stakeholders to manage crop health effectively and proactively. The expected outcome is a robust AI-powered decision support tool that empowers stakeholders to reduce crop loss, improve plant health, and optimize tea production. While the disease detection module is the centerpiece, the integration with grading, yield prediction, and market demand forecasting ensures a comprehensive platform that enhances overall productivity and sustainability in the Sri Lankan tea industry.

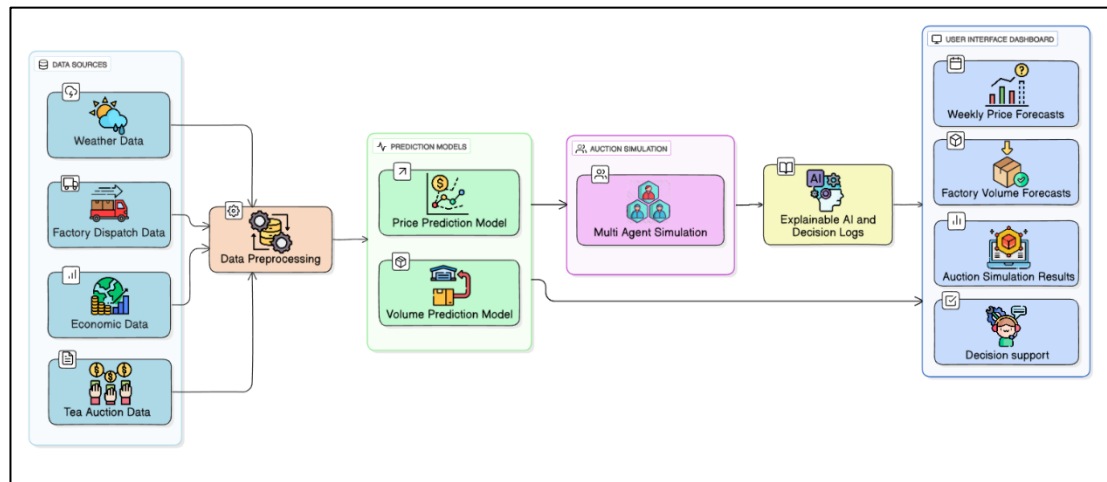


Figure 3.2: Tea Auction Forecasting and Simulation Framework

This diagram illustrates the complete workflow of the proposed system. Input data such as weather, production, and market exchange rates are first preprocessed and

modeled using SARIMAX for price prediction. A hybrid ARIMA–Random Forest approach is then applied for weekly price and volume forecasting. These forecasts feed into a Multi-Agent Simulation, which models the interactions between buyers, brokers, and factories, reflecting real auction dynamics. Finally, the system generates dashboards, metrics, and actionable recommendations, providing stakeholders with transparent insights into optimal selling, storing, or bidding strategies. This visualization emphasizes the integration of prediction and simulation to enhance decision-making in the Colombo Tea Auction.

3.2 Software Development Process

The development of the proposed web-based tea decision-support system follows an Agile Software Development methodology, emphasizing iterative progress, stakeholder collaboration, and rapid adaptability to evolving requirements. This approach allows continuous refinement of the forecasting models, multi-agent simulations, and user interface based on user feedback and system performance. Each development sprint focuses on implementing , testing specific features, such as weekly price prediction, volume forecasting, or auction simulation, ensuring that functional modules are incrementally integrated into the overall system. By adopting Agile practices, the project can respond efficiently to new insights from data analysis, address unforeseen challenges, and maintain alignment with the practical needs of stakeholders including tea brokers, factory managers, and smallholder farmers. The cycle of iterative development, review, and adjustment ensures the delivery of a reliable, user-centric platform with measurable improvements at each stage.

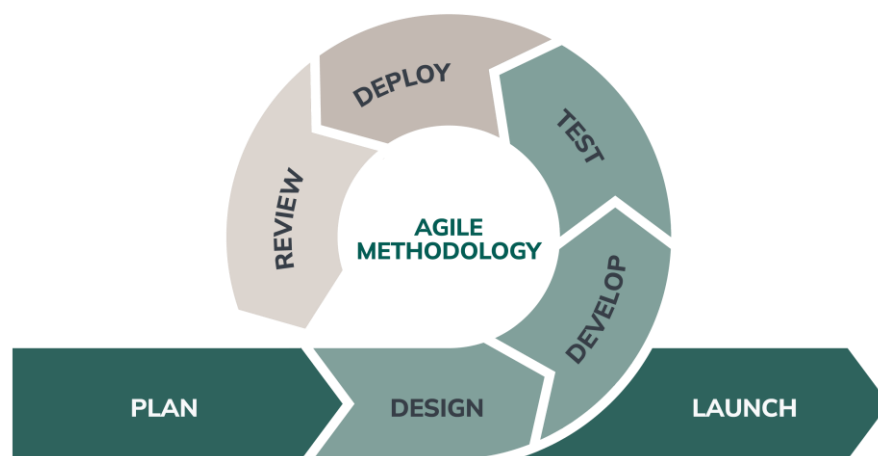


Figure 3.3: Software Development Life Cycle

3.3.1 Requirements Gathering and Analysis

As the first step of requirement analysis, a series of discussions were carried out with Dr. Rohitha, Senior Researcher at the Hanthana Tea Research Institute, and Mr. Nishan a representative from Colombo Tea Brokers Ltd., to gain a practical understanding of the challenges faced in the Sri Lankan tea market. These meetings provided crucial insights into the gaps in demand forecasting and auction planning, and helped to define the key functional requirements of the system.

Dr. Rohitha highlighted that Sri Lanka's tea production is highly sensitive to geographical elevation, climate variations, and seasonal factors, yet there is no systematic tool to integrate these variables into market predictions. He emphasized that production and supply often do not align with actual global demand, leading to volatility in tea prices and inefficiencies in the auction process.

From the broker's perspective, Mr. Nishan explained that while the Colombo Tea Auction remains the primary marketplace for Ceylon Tea, the absence of advanced forecasting tools limits the ability of both buyers and sellers to make data-driven decisions. He further noted that smaller producers, in particular, lack access to timely market insights, which places them at a disadvantage compared to larger traders. According to him, a system that can analyze historical auction records, global tea demand trends, and elevation-based production data would help mitigate risks and stabilize price fluctuations.

Based on these requirements, the decision was made to design a Tea Market Demand Forecasting and Smart Auction Planning system. This platform will leverage machine learning algorithms and predictive analytics to forecast demand patterns, while also providing auction planning support by suggesting optimal selling times, price expectations, and competitive strategies. The overall goal is to increase market transparency, improve price stability, and enhance the decision-making power of both large-scale exporters and smallholder tea producers.

3.3.2 Feasibility Study

• **Technical Feasibility**

All team members have prior experience in machine learning model development, data analysis, and web-based system design. The project requires building predictive models for tea market demand using historical auction data, production statistics, and global demand trends. In addition, the system will feature a web-based dashboard for visualization, along with database integration for storing and retrieving tea auction records, production data, and predictions. Cloud-based deployment is considered for scalability and real-time updates.

Table 3.1: Tools & Technologies

Category	Tools/Technologies
Programming Languages	Python, React
ML Frameworks	Scikit-learn, TensorFlow, PyTorch
Data Processing & Visualization	Pandas, NumPy, Matplotlib, Power BI
Algorithms	Regression Models, Time Series Forecasting (ARIMA, LSTM, SARIMAX), Clustering
Dataset Sources	Colombo Tea Auction Reports, Sri Lanka Tea Board, Hanthana Tea Research Institute

• **Economic Feasibility**

The system development is economically viable since most of the required tools and frameworks are open-source. The primary expenses involve data collection, cloud hosting, and maintenance. However, the potential benefits such as better price forecasting, optimized auction planning, and improved decision-making for brokers and producers far outweigh the costs. Moreover, the platform could be extended as a subscription-based service for tea brokers and exporters, ensuring a sustainable revenue model.

• Operational Feasibility

The system is designed to be practical and user-friendly, ensuring smooth adoption by its intended stakeholders such as tea brokers, exporters, producers, and policymakers. Since tea auctions in Sri Lanka are conducted weekly under a structured process, the integration of a forecasting and smart auction planning platform directly supports existing workflows without major disruption.

• Scheduling Feasibility

The proposed system can be completed within the allocated project timeline, as each phase (requirement gathering, dataset preparation, model training, web application development, testing, and deployment) will be carefully managed with time constraints. The Gantt chart outlines the schedule for each activity to ensure timely delivery of a quality product.

3.3.3 Dataset

The dataset for this research was compiled from multiple verified and authoritative sources to ensure high accuracy, consistency, and relevance to Sri Lanka's tea industry. The Colombo Tea Auction serves as the central marketplace for Ceylon Tea, and therefore, weekly auction reports from the Sri Lanka Tea Board form the foundation of this dataset. These reports provide granular information on auction dates, brokers, factory codes, grades, quantities, prices, and buyer participation details. Additionally, factory-level dispatch and production data were collected from 28 Mid-Country factories, providing detailed weekly dispatch volumes, production quantities, and tea grades categorized by elevation (Low, Mid, and High). These records enable factory-level analysis for the proposed volume forecasting model.

The dataset spans 2023–2024, comprising over 500,000 rows per year, making it one of the most comprehensive digital collections of Sri Lankan tea auction records. Each entry contains multiple dimensions broker, grade, buyer, factory, and elevation allowing precise modeling of weekly auction dynamics.

Screenshots of sample data tables are shown below:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
YEAR	MONTH	BROKER	LOT NO	SALE NO	INV NO	MF	REF	MARK	SALE TYPE	GRADE	QTY	PRICE	PRO	TYPE
2023	JAN	AS	1	1 414	MFH	344	GALEWELA SUPER	PUBLIC AUCTION	BOPF	1120	900	1008000	MAIN	
2023	JAN	AS	2	1 415	MFH	344	GALEWELA SUPER	PUBLIC AUCTION	BOPF	1120	920	1030400	MAIN	
2023	JAN	AS	4	1 66	MF	62	KELANEIYA & BRAEMAR	PUBLIC AUCTION	BOPF	580	1500	870000	MAIN	
2023	JAN	AS	5	1 248R	MF	1003	SHANNON	PUBLIC AUCTION	PEK	490	1350	648000	MAIN	
2023	JAN	AS	7	1 235R	MF	1003	SHANNON	PUBLIC AUCTION	BOPF	1160	1600	1856000	MAIN	
2023	JAN	AS	8	1 241R	MF	1003	SHANNON	PUBLIC AUCTION	BOPF	1160	1550	1798000	MAIN	
2023	JAN	AS	9	1 243R	MF	1003	SHANNON	PUBLIC AUCTION	BOPF	1160	1500	1740000	MAIN	
2023	JAN	AS	10	1 270R	MFA	890	THERESIA	PUBLIC AUCTION	BOPF	580	1550	899000	MAIN	
2023	JAN	AS	13	1 235R	MF	495	FORDYCE	PUBLIC AUCTION	BOPF	1120	1550	1736000	MAIN	
2023	JAN	AS	15	1 273R	MFA	981	ANNFIELD	PUBLIC AUCTION	BOP	1000	1600	1600000	MAIN	
2023	JAN	AS	16	1 270R	MFA	981	ANNFIELD	PUBLIC AUCTION	BOPF	1120	1650	1848000	MAIN	
2023	JAN	AS	17	1 275R	MFA	981	ANNFIELD	PUBLIC AUCTION	BOPF	1120	1750	1960000	MAIN	
2023	JAN	AS	18	1 279R	MFA	981	ANNFIELD	PUBLIC AUCTION	BOPF	1120	1650	1848000	MAIN	

TYPE	ELE	BUYER CODE	BUYER NAME	CHEST	20	EX-ESTATE	MF	MFREF	FACTORY	ELE1	ELE2	GROUP
MAIN		J2	A.F. JONES (EXPORTS) CE		20	EX-ESTATE	MF	MF0344	MATALE WEST	Medium	WM	0
MAIN		J2	A.F. JONES (EXPORTS) CE		20	EX-ESTATE	MF	MF0344	MATALE WEST	Medium	WM	0
MAIN		S080	SUNSHINE CONSUMER LANKA		10	EX-ESTATE	MF	MF0062	KELANEIYA	High	WH	0
MAIN		D332	DILMA CEYLON TEA CO PLC		10	EX-ESTATE	MF	MF1003	SHANNON	High	WH	0
MAIN		U102	UNI WORLD TEAS (PVT) LT		20	EX-ESTATE	MF	MF1003	SHANNON	High	WH	0
MAIN		A41	AKBAR BROTHERS LTD.		20	EX-ESTATE	MF	MF1003	SHANNON	High	WH	0
MAIN		U102	UNI WORLD TEAS (PVT) LT		20	EX-ESTATE	MF	MF1003	SHANNON	High	WH	0
MAIN		A41	AKBAR BROTHERS LTD.		10	EX-ESTATE	MF	MF0890	ST THERESIA	High	WH	0
MAIN		S080	SUNSHINE CONSUMER LANKA		20	EX-ESTATE	MF	MF0495	FORDYCE	High	WH	0
MAIN		A41	AKBAR BROTHERS LTD.		20	EX-ESTATE	MF	MF0981	ANNFIELD	High	WH	0
MAIN		U102	UNI WORLD TEAS (PVT) LT		20	EX-ESTATE	MF	MF0981	ANNFIELD	High	WH	0
MAIN		S080	SUNSHINE CONSUMER LANKA		20	EX-ESTATE	MF	MF0981	ANNFIELD	High	WH	0
MAIN		D332	DILMA CEYLON TEA CO PLC		20	EX-ESTATE	MF	MF0981	ANNFIELD	High	WH	0
MAIN		S080	SUNSHINE CONSUMER LANKA		20	EX-ESTATE	MF	MF0552	WANARAJAH	High	WH	0
MAIN		D332	DILMA CEYLON TEA CO PLC		20	EX-ESTATE	MF	MF0663	BOGAWANTALAWA	High	WH	0
MAIN		T43	TEA TANG (PVT) LTD.		20	EX-ESTATE	MF	MF0348	NORWOOD	High	WH	0
MAIN		T43	TEA TANG (PVT) LTD.		20	EX-ESTATE	MF	MF0348	NORWOOD	High	WH	0
MAIN		A41	AKBAR BROTHERS LTD.		20	EX-ESTATE	MF	MF0348	NORWOOD	High	WH	0

Figure 3.4: Auction Transaction Records (Factory, Grade, Quantity, Price, Buyer, Elevation, and Factory-Level Details)

3.3.4 Implementation

The implementation of the proposed Tea Market Forecasting and Smart Auction Planning system follows an Agile development approach, ensuring iterative development, regular testing, and stakeholder feedback. The implementation will be divided into four main modules:

- **Backend Development:** The backend of the system is developed using Python with FastAPI or Flask to provide RESTful APIs. It handles data ingestion, preprocessing, and storage in a centralized database. The backend also manages model execution and communication with the frontend, ensuring that users can access real-time predictions and simulations seamlessly.
- **Machine Learning Model Development:** The forecasting models form the core of the system. SARIMAX and LSTM are used for weekly tea price prediction, capturing seasonal and elevation-specific variations, while a hybrid ARIMA–Random Forest model is applied for factory-level volume forecasting. These models are trained and evaluated using performance metrics such as MAPE and RMSE to ensure accuracy and reliability.
- **Recommendation Generation:** The system provides actionable recommendations based on model outputs and simulation results. These include suggestions on the best time to sell, expected auction risks, and storage decisions. Explainable AI (XAI) methods such as SHAP and LIME are integrated to make the recommendations transparent, showing users how different factors influenced the predictions.

- **Web Application Development:** The frontend is implemented using frontend frame work to deliver an interactive and user-friendly interface. Dashboards and charts display price forecasts, volume trends, and auction outcomes, while the Multi-Agent Simulation (MAS) module allows users to test different market scenarios. Results are visualized with clear explanations, enabling stakeholders to make informed and confident decisions.

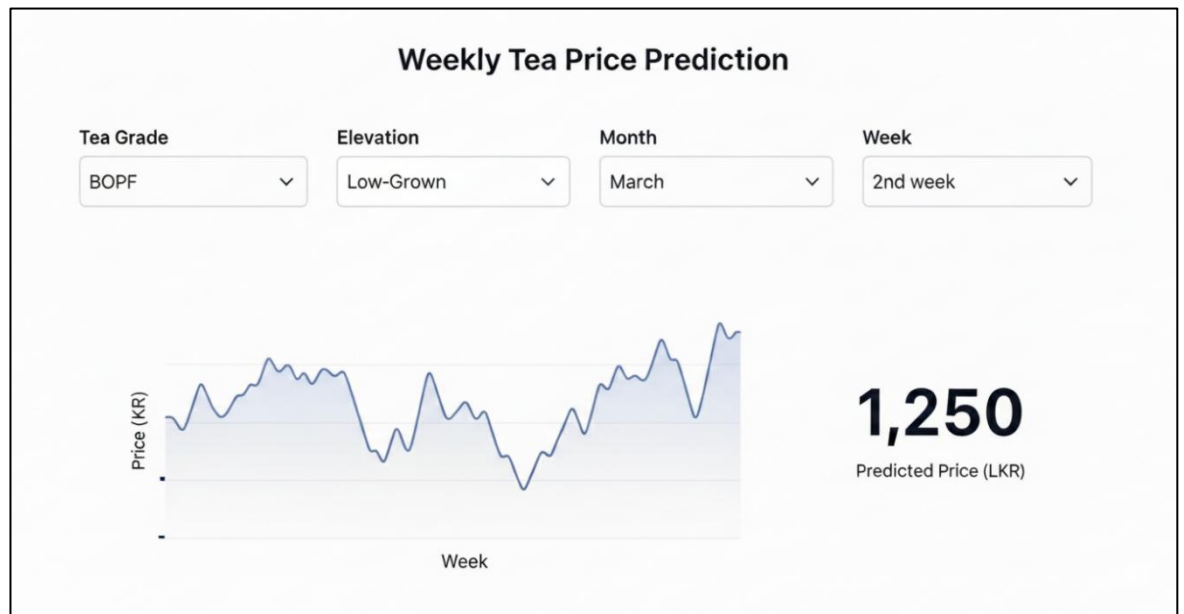


Figure 3.5: Wireframe of Tea Price Forecasting Interface

This wireframe shows how users can select the tea type, elevation, month, and week to view the predicted price. A chart is also displayed to show how prices are expected to change during that month. This helps users easily understand weekly price trends and decide the best time to sell.

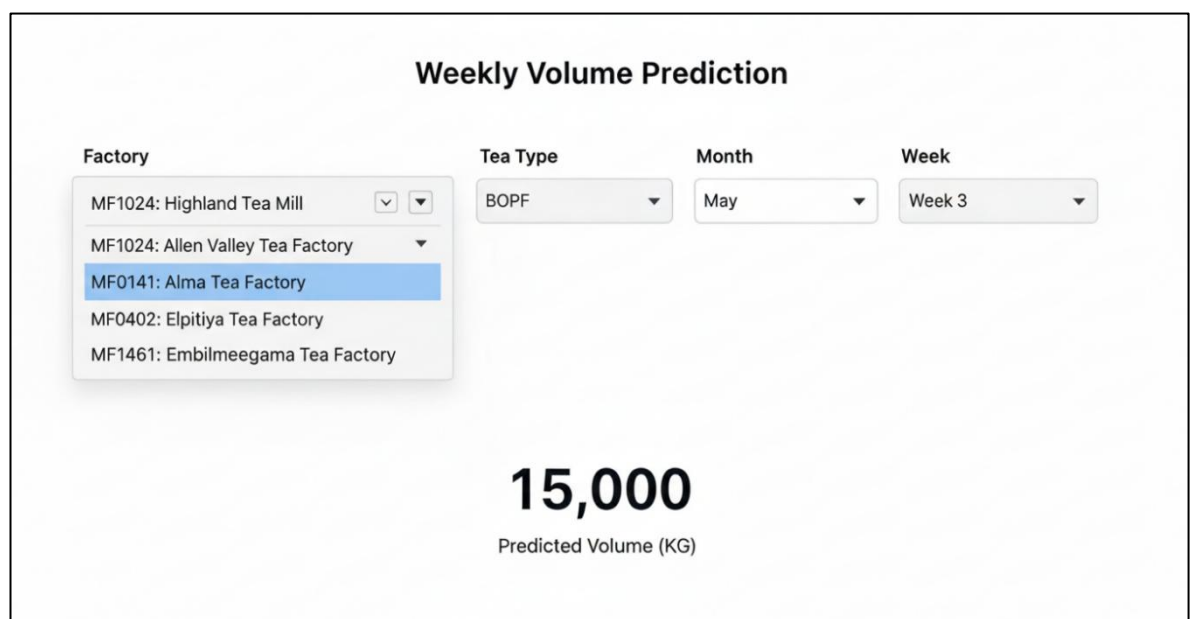


Figure 3.6: Wireframe of Factory-Level Volume Prediction Interface

This wireframe allows the user to select any Mid-Country tea factory (out of 28), choose the tea type, month, and week, and view the predicted auction volume for that period. It helps factory managers plan how much tea to send to the auction each week, reducing the risk of oversupply or shortages.

The wireframe is divided into three main sections: User Input Panel (Selections), Simulation Outcome, and Explainability (XAI Layer).

User Input Panel (Selections)

- Factory Parameters**
 - Weekly volume 550: 30000 ✓
 - Minimum acceptable price (Rs/kg): ✓
- Buyer Parameters**
 - Number of buyers: 1 Dust + ✓
 - Budget range (Sc.):
 - Storage cost week: 1M - 3M
- Market Parameters**
 - Plade preference: ✓
 - Demand level: Medium
 - Competition intensity:
 - Festival: Weak
 - Seasonal Normal week

Run Simulation

Simulation Outcome

- Avg. Price**
Rs. 1,020 / kg ✓
- 100% volume sold**
- Buyer 2 won most lots

Explainability (XAI Layer)

- Factory A** ✓
"I sold all 1000kg because xtte... he festival increesle cōmpnyuyor Rs. 950/kg nūgin of toes incrch. TEL. ✓
- Buyer 2** ✓
I outbid ode hers beccause Rs. 2.5M. I prioritizzed pridges as Rs. BOPF grade tvpstivahad resale prices."
- Broker:**
Average auction price rose by 8% compared us higher competition. The increased to volumye abyorseuolag. The market clearrine sold and ouscessfully overssupply.
- System Summary**
Overall market demand. High
• Competition intensity. Weak
Optimal strategy. Increase supply >

Figure 3.7: Wireframe of Auction Simulation Interface

This wireframe shows the Smart Auction Simulation. Users can enter factory details (volume, minimum price, storage cost), buyer details (number of buyers, budgets, grade preferences), and market settings (demand level, competition, seasonal factor). The simulation engine (MAS) then models the auction with factory, buyer, and broker agents. Results such as average price, sold volume, and buyer allocations are displayed, together with explanations (XAI) that describe why outcomes occurred. This design helps stakeholders test “what-if” scenarios (e.g., festival demand vs.

normal weeks) and understand both predictions and auction behaviors before real trading.

3.3.5 Testing

Testing is carried out after the implementation phase and before the final deployment of the proposed system. It is a critical stage of the Software Development Life Cycle (SDLC), ensuring that the system is reliable, accurate, and user-friendly. Testing validates that all modules function as intended and that the overall platform meets both functional and non-functional requirements.

- **Unit Testing** – Each module (price prediction, volume forecasting, auction simulation) is tested independently by the development team to verify its correctness. Unit testing ensures that individual components such as data input, prediction generation, and visualization work as expected.
- **Integration Testing** – Once unit testing is complete, all modules are integrated and tested together. This step ensures that the forecasting models, simulation engine, and user interface communicate smoothly without errors. For example, the predicted price outputs are validated to be correctly passed to the auction simulation module.
- **System Testing** – The entire web-based decision-support system is tested in a real-world environment. This includes validating accuracy of predictions, running multiple auction scenarios, and testing the responsiveness of the user interface. System testing guarantees that the application functions as a whole and fulfills its intended purpose.

3.3.6 Work Breakdown Structure

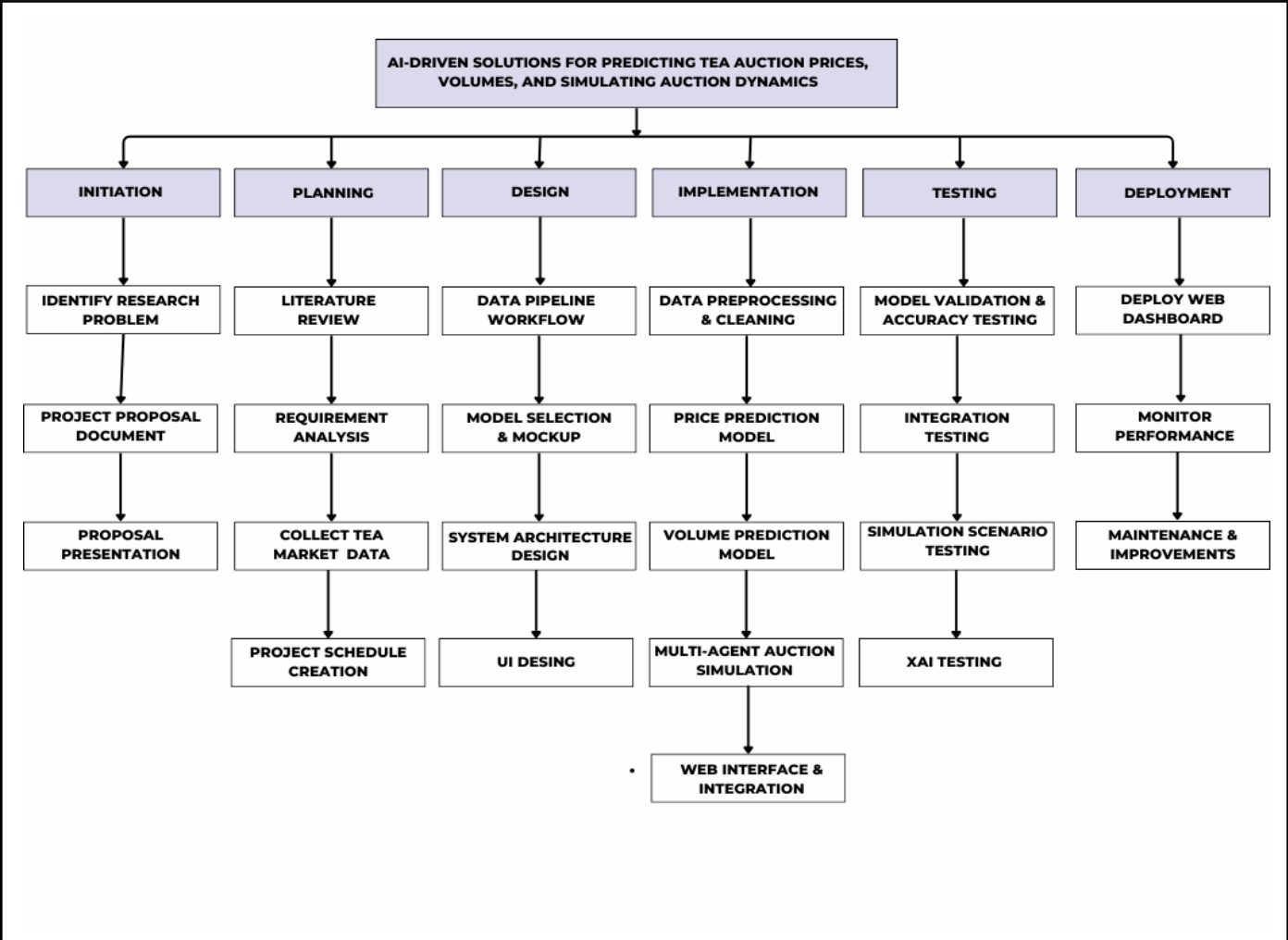


Figure 3.8: Work Breakdown Structure

3.3.7 Gantt Chart

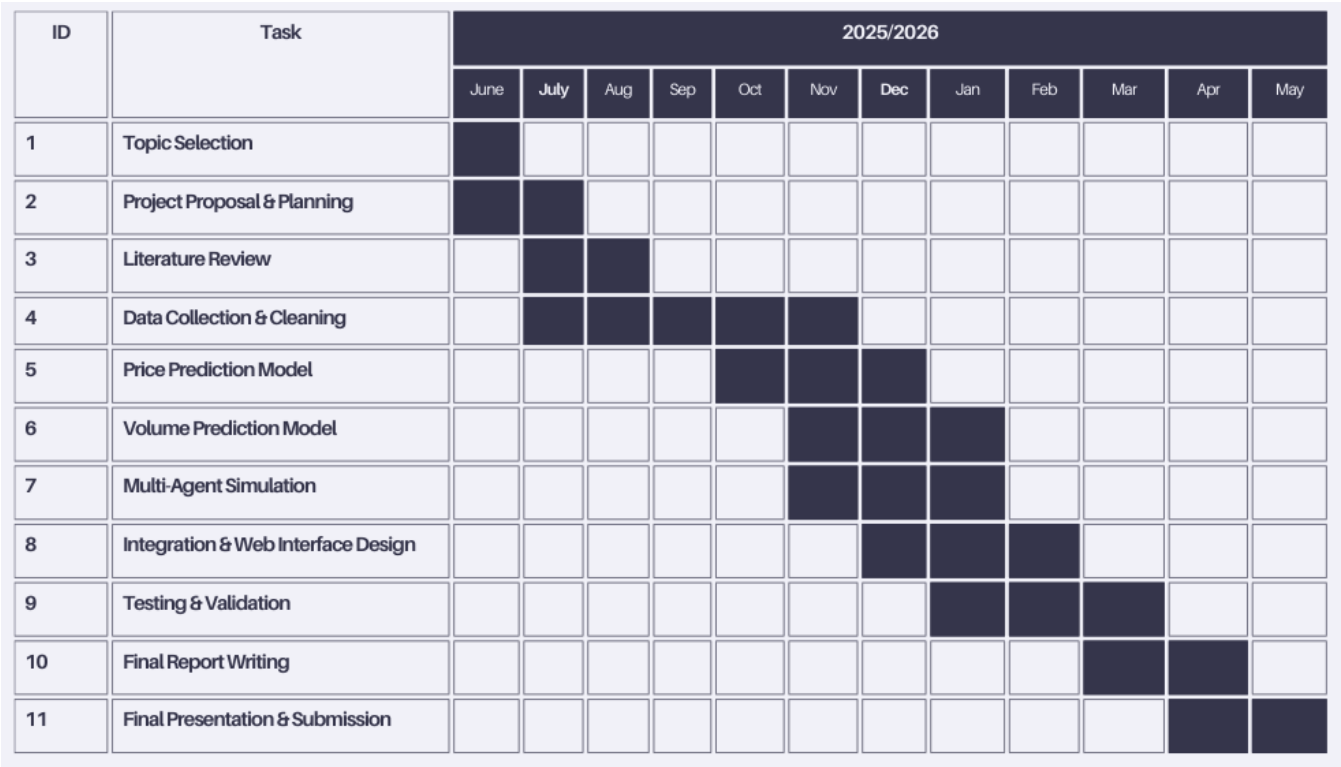


Figure 3.9: Gantt Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- Forecast market demand – The system should analyze historical tea auction data, production figures, and external factors (such as exchange rates) to forecast future market demand trends.
- Simulate smart auction planning – The system should allow stakeholders (factories, brokers, buyers) to run auction simulations, predicting price fluctuations and buyer demand behavior before actual trading.
- View demand and price trends – Users should be able to visualize tea price patterns, demand variations, and correlations with external factors through interactive charts and dashboards.
- Generate reports – The platform should automatically generate detailed reports on demand forecasts, auction outcomes, and performance trends for stakeholders.
- Provide real-time insights – The system should provide up-to-date analytics during live auctions to support data-driven decision-making.
- Enable multi-user roles – The system should support different user roles (factories, brokers, buyers, and administrators), each with tailored functionalities according to their needs.

4.2 Non-Functional Requirements

When developing this system, it is essential to take non-functional requirements into account as well. Since our target users include tea factories, brokers, buyers, and administrators, the system must provide secure, reliable, and insightful tools to support market forecasting and smart auction planning. Beyond this, there are several other important non-functional attributes that must be considered, as outlined below.

- Availability – The platform should be accessible at all times, especially during critical auction periods, without interruptions or downtime

- **Reliability** – The system must consistently deliver accurate demand forecasts and auction simulations, ensuring stakeholders can rely on the insights for decision-making.
- **User-Friendly Interface** – The interface should be simple and intuitive, presenting data visualizations, forecasts, and simulation outcomes in a way that is easy for non-technical users to understand.
- **Performance** – The system should handle large datasets (auction records, production data, and external market indicators) efficiently and provide results within seconds to support real-time decisions.
- **Security** – Sensitive data such as factory production figures, buyer demand, and auction outcomes must be protected with secure authentication, encryption, and access control.
- **Scalability** – The solution should be able to grow with increasing numbers of users and larger datasets without performance degradation.
- **Cost-Effective** – The system should be designed and implemented with efficient use of resources, ensuring it remains affordable and practical for the tea industry stakeholders.

4.3 User Requirements

1. The user should be able to input tea production and auction-related data.
2. The user should be able to view demand forecasts in graphical and tabular formats.
3. The user should be able to simulate smart auction scenarios and observe predicted price trends.
4. The user should be able to generate and download reports of forecasts and auction outcomes.
5. The user should be able to access real-time insights during live auction sessions.
6. The user should be able to switch between user roles (factory, broker, buyer, admin) with role-specific features.

4.4 Software Requirement

User-End

- Web Browser - The application should be accessible through modern browsers such as Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge.

Developer-End

- IDEs / Tools -Microsoft Visual Studio Code, Workbench
- Programming Languages -Python or above (backend, ML models), Frontend Framework (frontend)
- Frameworks - Flask or above (for RESTful API development), Machine Learning frameworks for forecasting and simulation
- Libraries - TensorFlow or above, Pandas, NumPy, Matplotlib/Plotly (for data visualization), Scikit-learn
- Version Control - Git and GitHub for collaborative development and version tracking

4.5 Expected Test Cases

Test case ID: SID_Test_01			Test designed by: Bandaranayake B.G.W.T.N. Test designed day: 2025/8/24		
Test title: Demand Prediction Accuracy					
Test priority (High/Medium/Low): High					
Module name: Demand Prediction Accuracy					
Description: Validate demand prediction for upcoming auction cycles					
Pre-conditions: Availability of historical production and sales data					
Testing steps: 1. Input historical production and sales data 2. Run demand prediction model					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
SID_Test_01	Historical tea production/sales data	Predicted demand matches expected trends (<5% error).	Predicted demand within expected error range.	Pass	Confirms demand forecast reliability

Test case ID: SID_Test_02			Test designed by: Bandaranayake B.G.W.T.N. Test designed day: 2025/8/24		
Test title: Auction Price & Volume Recommendation					
Test priority (High/Medium/Low): High					
Module name: Smart Auction Planning					
Description: Validate system recommendations for auction prices and volumes					
Pre-conditions: Demand prediction results available.					
Testing steps: 1. Input forecasted demand and past auction data 2. Run auction planning module					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
SID_Test_02	Forecasted demand + past auction data	Suggested price within realistic market range	Suggested price within competitive range	Pass	Validates pricing reliability
SID_Test_02	Factory dispatch volume	Auction volume avoids overstock/shortage (balanced)	Auction volume matched predicted balance	Pass	Confirms market-demand alignment

Test case ID: SID_Test_03			Test designed by: Bandaranayake B.G.W.T.N.		
Test title: Smart Auction Summarization Accuracy			Test designed day: 2025/8/24		
Test priority (High/Medium/Low): High					
Module name: Smart Auction Planning					
Description: Validate that the system generates an accurate and concise summary of tea auction results (prices, demand trends, and factory participation).					
Pre-conditions: Auction data for the selected week is available in the system.					
Testing steps: 1. Select a completed tea auction session from the auction history. 2. Click on “Run Auction Sumulization .” 3. Verify generated summary for correctness and clarity.					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
SID_Test_03	Weekly auction data (lots, bids, prices)	Summary with key stats (avg. price, highest bid, top-demand teas, factory trends)	Summary displayed as expected	Pass	S ummary provides clear auction insights
SID_Test_03	Incomplete auction data (missing bids)	Error or partial summary with warning	Warning displayed, partial summary	Pass	Handles missing data gracefully

5. BUDGET AND BUDGET JUSTIFICATION

The estimated budget for this project mainly covers travel expenses and website hosting charges. Traveling costs are necessary for visiting relevant institutions such as tea factories, the Sri Lanka Tea Board, or research centers to collect historical data and conduct validation discussions. In addition, a portion of the budget is allocated for website hosting, which is required to deploy the forecasting and auction simulation system online for stakeholder access. Since the system relies primarily on open-source tools and locally available resources, no significant costs are expected for cloud subscriptions or external devices.

Table 5.1: Budget of the Project

Component	Amount (USD)	Amount (LKR)
Traveling expenses for data collection and consultations (per month)	11.67	3,500.00
Website hosting charges (per month)	16.00	4,800.00
Total	27.67	8,300.00

6. COMMERCIALIZATION

The commercialization strategy of the Tea Market Demand Forecasting and Smart Auction Planning System focuses on empowering tea brokers, factory managers, exporters, and research institutions with accurate forecasting and intelligent auction planning tools. By introducing a subscription-based model tailored to brokers, factories, and enterprises, the system ensures affordability while delivering high-value insights that minimize risks of oversupply, underpricing, and missed export opportunities. Partnerships with industry stakeholders such as the Sri Lanka Tea Board, Colombo Tea Brokers, and the Hantana Tea Research Institute will be key to building trust and widespread adoption, while marketing efforts will utilize professional networks, industry forums, and digital platforms to reach target users. In the long run, the system aims to establish itself as a core decision-support tool within Sri Lanka's tea industry and expand regionally, ensuring sustainability and continued improvements based on user feedback and evolving market demands.

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